Addressing Mode

Computer Architecture and Organization

Addressing Modes

- The operation field of an instruction specifies the operation to be performed. This operation must be executed on some data stored in computer registers or memory words.

- The way the operands are chosen during program execution is dependent on the addressing mode of the instruction.

- The addressing mode specifies a rule for interpreting or modifying the address field of the instruction before the operand is actually referenced.

Program Counter

- The PC register keeps track of the instructions in the program stored in memory.

- PC holds the address of the instruction to be executed *next* and is incremented each time an instruction is fetched from memory.

Categories of Addressing Mode

The most well known addressing mode are:

- Implied Addressing Mode.
- Immediate Addressing Mode
- Register Addressing Mode
- Register Indirect Addressing Mode
- Auto-increment or Auto-decrement Addressing Mode
- Direct Addressing Mode · Indirect Addressing Mode
- Displacement Address Addressing Mode
- Relative Addressing Mode · Index Addressing Mode

Example to be used in the following sections :-

PC=200

R1=400

XR=100

AC

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Direct Address

- It is the address (in memory) of the data to be used: *“address of the operand”* .
- Effective address is 500 and the operand to be loaded into AC is 800.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Indirect Address

- It is the address(in memory) of the *address of the data* to be used.
- Effective address is 800 and the operand is 300.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Register Mode

- Operands are in registers that reside within the CPU.
- R1=400

  Simply, the operand is in R1 and 400 is loaded into AC.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Register Indirect Mode

- In this mode, the selected register contains the address of the operand rather than the operand itself.
- R1=400

  Effective address is 400, equal to the content of R1 and the operand loaded into AC is 700.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Immediate Mode

- In this, operand is specified in the instruction itself. It has an operand field rather then the address field.
- Operand field contains the actual operand to be used in conjunction with the operation specified in the instruction.
- The second part of the instruction is taken as an operand rather than an address

So, 500 is loaded into AC.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Auto increment Mode

- This is similar to register indirect mode except that the register is incremented after its value is used to access memory.
- R1=400, R1 is incremented i.e. R1=401 after the execution of the instruction.

Operand=700

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Auto decrement Mode

- In this mode, the register is decremented before its value is used to access memory.
- R1=400, R1 is decremented i.e. R1=399

The operand loaded into AC is now 450.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Relative Addressing Mode

- In this mode, the content of PC is added to the address part of the instruction in order to obtain the effective address.
- The effective address is 500+202=702 and the operand is 325.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Indexed Addressing Mode

- In this mode the content of an index register is added to the address part of the instruction to obtain the effective address.

•XR=100

XR+500=100+500=600

Operand= 900

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Sample Question

- Registers *R1* and *R2* of a computer contain the decimal values 1200 and 5600. What is the effective address of the memory operand in each of the following instructions?
- Load ***20(R1), R5***
- Move ***#3000, R5***
- Store ***R2, R5***
- Add ***–(R2), R5***
- Subtract ***(R1)+, R5***
- Add ***R5, 30(R1,R2)***

Thank You!